BRANCH 03

Cross-Functional Collaboration

Boundaries, shared understanding, and the work of integration

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Overview

Cross-functional work is often described as a meeting of perspectives. More precisely, it is a meeting of interdependencies: one group's choice changes another group's costs, timing, risks, or room to maneuver. Expertise makes this work possible, but expertise also gives each function its own vocabulary, evidence standards, time horizon, and account of what counts as success. Collaboration therefore requires more than congenial relationships. It requires structures through which people can discover dependencies, translate consequences, locate relevant knowledge, and revise a course of action while preserving dignity and accountability.

The three essays in this reader approach that task from different angles. The first offers an integrative account of how shared purpose, boundary objects, reciprocal communication, and learning loops can convert specialist difference into coordinated action. The second examines the limits of exhortation when incentives, status, workload, and decision rights pull functions apart. The third moves to the human scale of the handoff, where organizational design becomes a lived exchange between people. Across all three, the central claim is cautious but consequential: high-quality collaboration is a disciplined capacity to keep difference useful while making mutual dependence visible, discussable, and fair.

ESSAY ONE

The Work Between Functions

Difference Is the Asset and the Difficulty

Organizations divide work into functions because specialization creates depth. Finance becomes fluent in exposure and return; operations in flow and reliability; sales in customers and commitments; technology in architectures and failure modes. The value of a cross-functional group lies in bringing these locally developed forms of knowledge to a problem that requires several functions. Yet the same division that creates expertise also produces partial views. A launch date may appear to marketing as a promise, to engineering as a probability distribution, to compliance as a threshold, and to service as a future queue. The disagreement may arise from a collision among legitimate representations of the work.

Cronin and Weingart (2007) call such differences “representational gaps”: people may believe they are debating solutions while actually working from different descriptions of the problem. In an ethnographic study of product development, Carlile (2002) similarly found that knowledge is localized, embedded in practice, and invested with consequences. Translation becomes harder when a proposed change redistributes effort or risk. These findings support treating alignment as an unfolding process. A team must first surface what each function sees, what it stands to gain or lose, and which dependency is being negotiated.

This approach preserves specialization. Coordinating expertise depends partly on knowing where knowledge resides and bringing it to bear at the right time (Faraj & Sproull, 2000). Transactive memory research describes a related capacity: effective groups develop credible knowledge of “who knows what,” allowing members to specialize while remaining connected (Lewis, 2003). The practical aim is sufficient mutual legibility. Each contributor should be able to explain both a recommendation and the constraint, uncertainty, or downstream consequence that makes it reasonable.

Give Interdependence a Visible Form

Abstract appeals to “work as one team” rarely show people where their work touches. A shared artifact can. A service blueprint, dependency map, decision log, customer journey, prototype, risk register, or scenario is useful when it gives different specialists something they can inspect and change together. Carlile (2002) describes such boundary objects as means of representing, learning about, and transforming knowledge across domains. Their value lies in locating ambiguity in a form that people can question together.

A good boundary object preserves consequential difference. A project plan that displays only milestones may hide uncertainty; a dashboard that collapses all risk into a single color may imply consensus while views remain divided. By contrast, an artifact can show confidence ranges, ownership, assumptions, affected groups, and unresolved choices. It can make clear that a “small” feature request creates substantial testing work, or that a technically elegant solution places an unreasonable burden on frontline staff. The object becomes a place where care is operational: people can see who absorbs the cost of a decision and invite those people into the decision before commitment hardens.

Coordination research suggests that plans, routines, meetings, and schedules work by supporting accountability, predictability, and common understanding (Okhuysen & Bechky, 2009). Those conditions are more informative than a preference for any particular tool. A daily meeting that leaves ownership diffuse is ceremony. A modest shared document that reveals a hidden dependency may be transformative. The question to ask of any coordinating mechanism is simple: what greater capacity to anticipate, understand, or answer for does it create?

Build a Reciprocal Communication System

Information sharing is positively associated with team performance, but the evidence also suggests that sharing is most valuable when information is uniquely held and relevant to the task (Mesmer-Magnus & DeChurch, 2009). The better goal is communication that is timely, selective, and reciprocal. A sender should state what changed, why it matters, what remains uncertain, and what response is needed. A receiver should be able to paraphrase the consequence, challenge an assumption, and signal the conditions required for action.

Reciprocity also depends on interpersonal conditions. Edmondson’s (1999) research linked psychological safety-shared confidence that interpersonal risk is tolerable-with learning behavior in work teams. It combines meaningful standards with practical permission to ask a basic question, disclose an error, or

oppose a premature consensus while preserving standing and dignity. Leader inclusiveness matters especially where professional status differs; in health-care teams, words and actions that invited participation helped reduce the silencing effects of status (Nembhard & Edmondson, 2006).

The design implication is to distribute voice intentionally. Ask the function that will inherit the risk to speak before the function with the greatest formal authority. Separate a round of problem description from solution advocacy. Invite written input before a meeting when rapid conversation favors fluent or senior participants. Record dissent and the conditions under which a decision should be revisited. These measures improve the conditions for candor while retaliation remains a plausible constraint, and they make reciprocal communication less dependent on individual courage.

Learn Through Intermediate Commitments

Novel work rarely yields to complete analysis in advance. Majchrzak, More, and Faraj (2012) observed cross-functional teams that generated intermediate scaffolds-provisional ideas that participants repeatedly modified-while some differences in specialist knowledge remained unresolved. Their finding complicates the assumption that deep mutual translation must always precede action. Sometimes a small, revisable object lets people learn enough from one another to keep creating.

This is the proper role of incremental progress. A pilot, mock-up, limited release, or rehearsal functions as both a bounded deliverable and a question posed to reality. The answer should return quickly to the people whose work and judgment shaped the trial. Team-cognition research finds meaningful relationships between shared cognitive structures, team processes, and performance (DeChurch & Mesmer-Magnus, 2010), while shared understanding remains continuously updated through action. Each cycle can clarify the problem, revise the map of expertise, and strengthen collective confidence through evidence of what the group can actually accomplish.

The scale of the commitment matters. Excessive scale invites defense of sunk costs; trivial scale yields little learning. A useful intermediate commitment is bounded, observable, safe enough to revise, and close to the dependency under dispute. It carries a named hypothesis: if we change this handoff, what should improve, for whom, and what unintended burden should we watch?

Sustain the Whole System

Cross-functional success can still be locally extractive. A project may meet its deadline because one coordinator absorbs after-hours translation, one operational group takes on invisible maintenance, or one junior member continually repairs relationships among senior colleagues. Boundary-spanning work brings information and resources across group lines, but it also consumes attention and can expose people to conflicting demands (Marrone, 2010). In fluid, overlapping teams, participation itself becomes dynamic and boundaries less clear (Mortensen & Haas, 2018). Explicit protection gives dependable connectors sustainable roles, capacity, and backup.

Sustainable collaboration therefore includes capacity accounting. The team should name translation, liaison, and repair as work; rotate it where feasible; build redundancy around key relationships; and examine whose core responsibilities are displaced by shared initiatives. Interteam coordination is associated with later project outcomes in longitudinal research, especially when established early (Hoegl et al., 2004), and its resource demands require time, authority, and recognition.

At its best, cross-functional work does more than combine inputs. It creates a dependable social and cognitive infrastructure: people know whom to approach, how to express consequential difference, what evidence will change a decision, and how to repair a miss. Gittell's (2002) research on relational coordination emphasizes shared goals, shared knowledge, mutual respect, and timely problem-solving communication. Those qualities preserve specialist boundaries while making the power and effort involved in crossing them explicit.

ESSAY TWO

Systems That Make Collaboration Possible

Align Local Incentives With Collective Outcomes

A familiar organizational drama begins with leaders asking functions to collaborate while measuring them in ways that reward separation. Procurement is praised for lowering unit cost, operations for stability, product teams for speed, and sales for booked revenue. Each group may behave rationally within its scorecard and still create a fragmented customer experience. When the resulting conflict is diagnosed as a "silo mentality," an institutional contradiction is recast as an attitude problem.

Cross-functional tension often contains genuine differences in knowledge, but it also reflects incentives, scarce resources, and the distribution of consequences. Carlile (2002) showed that knowledge boundaries become especially difficult when interests are at stake. A request carries both information and a potential transfer of budget, reputational exposure, or late work. Cronin and Weingart's (2007) account of representational gaps helps explain why well-intentioned specialists can pursue incompatible versions of success and invites explicit attention to who has authority to impose a version on others.

The first task of collaboration is therefore diagnostic. Which outcomes are jointly owned? Which costs remain invisible because they fall outside the sponsoring function's measures? Which deadlines became promises ahead of participation by those expected to deliver them? A shared goal becomes credible when governance and rewards make attention to that goal possible. Calls for partnership then rest on a design that supports collective action.

Voice Is Shaped by Status and Consequence

Psychological safety is frequently invoked as though leaders can create it by welcoming questions. Edmondson (1999) defined and studied it as a group-level belief related to interpersonal risk that extends beyond friendliness. The risk varies with status. A senior expert may experience disagreement as spirited debate; a contractor, newcomer, or employee from a marginalized group may reasonably calculate that the same act threatens access, evaluation, or continued employment.

Nembhard and Edmondson (2006) found that leader inclusiveness could support engagement in improvement work across professional-status differences. Yet an invitation to speak is weak if decision processes consistently disregard the invited knowledge. A credible voice system closes the loop: it states who will decide, records how input affected the choice, explains the final judgment, and protects those who surface unwelcome facts. It also recognizes that silence may reflect learned caution alongside a wealth of unspoken ideas.

Communication protocols can help, though their reach remains limited against retaliation, employment insecurity, or unequal access to influential networks. When leaders ask for candor, they incur obligations: to protect people from punishment, distinguish challenge from disloyalty, and share the reputational cost of raising a problem. Care becomes inclusion when participation has a route to consequence.

The Unequal Burden of Bridging

Boundary spanning is often praised as a leadership capability. Marrone's (2010) review describes activities such as obtaining resources, representing a team, and coordinating tasks across its edges. These activities can improve access and adaptation, but they also create role strain. Connectors translate vocabulary, anticipate objections, maintain relationships, and soothe conflict-labor that is cognitively and emotionally demanding despite its frequent absence from project plans.

Organizations often allocate this work informally to those perceived as agreeable, socially skilled, or broadly connected. The result can be a hidden tax on people already expected to perform belonging for others. The same person is asked to mentor newcomers, interpret between functions, remember stakeholder histories, and rescue broken handoffs, all while being evaluated on individual output. Collaboration built on voluntary repair by a few conscientious employees becomes fragile and unfair.

Fluid participation intensifies the problem. Mortensen and Haas (2018) argue that contemporary teams are often overlapping and loosely bounded. Under these conditions, calendars fill with coordination while the architecture of coordination often lacks clear ownership. The organization benefits from flexible access to expertise, while individuals carry switching costs and ambiguous obligations. Goodwill may delay failure, yet finite attention also requires capacity and prioritization.

Decision Rights Are a Form of Communication

Many apparent communication failures are unresolved questions of authority. People exchange updates repeatedly because ownership remains unclear for trading schedule against scope, accepting risk, or sending a local choice to system-level review. Okhuysen and Bechky (2009) identify accountability, predictability, and common understanding as integrating conditions of coordination. Clear decision rights contribute to all three. They tell participants who acts, whose knowledge must enter before action, and where an appeal can go.

This clarity should preserve distributed authority. Expertise coordination works when relevant knowledge is recognized and mobilized (Faraj & Sproull, 2000). A sound arrangement places routine choices near the work while reserving genuinely cross-system tradeoffs for a forum with the authority and information to make them. It defines thresholds for escalation, time limits for consultation, and conditions that reopen a decision. In other words, governance itself becomes a feedback system.

Shared artifacts can expose whether governance is real. A decision record that names affected groups, consulted expertise, assumptions, and review dates makes omission visible. A dependency map can show when one function receives risk in excess of its resources. These tools gain value by creating institutional memory and reducing the ease with which power can rewrite how a commitment was made.

Redesign Before You Rally

Evidence on teams supports investments in information sharing, shared cognition, and relational coordination (DeChurch & Mesmer-Magnus, 2010; Gittell, 2002; Mesmer-Magnus & DeChurch, 2009). These relational and cognitive resources become durable when capacity, incentives, and authority align with them. Effective change revises bonus systems that reward unilateral commitments, supplies enough staffing for response, and treats each group's sacrifices and constraints as part of the shared purpose.

System repair begins by tracing recurrent friction to its production conditions. If every launch produces a late compliance crisis, move compliance knowledge and decision gates earlier. If operations repeatedly inherit unsupported features, give operational cost standing in portfolio choices. If a single liaison holds the partnership together, create role clarity, backup, and workload credit. If joint work succeeds only through exceptions, revise the rule.

The deepest form of collaboration is institutional honesty: acknowledging interdependence in the way work is funded, measured, scheduled, and governed. Every design meets unforeseen contingencies, so relationships remain essential. When the system carries its share of the coordinating burden, people can direct trust toward learning and invention as the organization resolves its contradictions.

ESSAY THREE

The Handoff as a Place of Learning

Every Transfer Contains a Theory

A handoff may look like a file sent, a ticket moved, a patient transferred, or a customer introduced. Beneath that visible act lies a theory: this information will be sufficient; the next person will interpret it as intended; the recipient has capacity; the work is ready; the exception is obvious. Most handoff failures arise when one of these assumptions remains private.

Specialists encode work in forms that make sense inside their own practice. Carlile (2002) found that knowledge at functional boundaries is localized and consequential. The sender who says “approved” may mean conceptually acceptable, while the receiver hears ready for execution. A small difference in language becomes a large difference in obligation. Treating the handoff as a learning site means asking what the transfer reveals about the relationship between two practices and how the next exchange can improve.

The most useful question after a miss is therefore, “What information or challenge pathway did a person in their position require?” That shift preserves accountability while opening the possibility of redesign. It uses the current exchange to improve the next handoff.

A Complete Signal Invites a Return Signal

Good handoffs carry more than content. They carry context, uncertainty, and a request. The sender identifies what changed, what has already been decided, what remains open, where the principal risk sits, and what response is expected by when. The receiver returns a signal: an acknowledgment, a paraphrase of the implication, a question, or a refusal with reasons. Communication becomes a closed loop that confirms shared understanding and next action.

Meta-analytic evidence connects information sharing with team performance, particularly where unique information reaches the group (Mesmer-Magnus & DeChurch, 2009). Yet volume can conceal the consequential signal. Copying more people into a thread may distribute anxiety while leaving responsibility

diffuse. A concise signal retains nuance and shapes it around the receiver's decision.

Reciprocity requires enough safety to expose uncertainty. Edmondson (1999) linked psychological safety with team learning behavior, including seeking feedback and discussing errors. In a handoff, this can be as modest as making "I need clarification on the acceptance criterion" a professionally legitimate response. Leaders help by rewarding early clarification and distinguishing fluent certainty from competence.

Let an Artifact Hold the Shared Question

When two functions repeatedly misunderstand each other, asking individuals to "communicate better" leaves translation in memory. A boundary object can hold the shared question outside either person's head. It might be a marked-up prototype, a checklist with exception fields, a sample customer record, or a decision page that distinguishes facts, assumptions, and open choices. The artifact should be detailed enough to reveal consequences and light enough to change.

Majchrzak, More, and Faraj (2012) describe how cross-functional teams used intermediate scaffolds to sustain creative engagement while some knowledge differences remained unresolved. This insight is humane as well as efficient. People can bring provisional disciplinary knowledge into a shared object, then point, revise, and learn together.

The object should also preserve authorship and affected voices. If a service representative identifies a recurring customer burden, the record should carry that observation into the design decision. If a junior analyst raises a data-quality concern, the concern should travel with the conditions needed to investigate it. Practical inclusion means that insight travels across status differences.

Repair Is a Capability

Every handoff system encounters surprise. The more consequential capacity is repair: recognizing a break early, containing harm, restoring mutual understanding, and changing the conditions that made recurrence likely. Gittell's (2002) work on relational coordination emphasizes timely, accurate, problem-solving communication supported by shared goals, shared knowledge, and mutual respect. Those qualities are tested most clearly when work goes wrong.

A repair conversation can begin with observable sequence and reserve motive for careful inquiry: what each person saw, what each believed would happen next, and where those accounts diverged. It can then identify an adjustment at three levels. The immediate task needs recovery. The relationship may need acknowledgment, especially if someone bore an avoidable cost. The system needs a changed cue, threshold, role, or artifact. An apology restores warmth; reliability also requires process redesign, and repair should name any injury.

Small repairs accumulate into collective efficacy: shared confidence in the group's capacity for coordinated action (Bandura, 2000). Lewis's (2003) research on transactive memory suggests that credibility, coordination, and specialization form a system through which groups use distributed knowledge. Each successful clarification teaches who knows what and demonstrates that asking for help produces movement with dignity. Each visible process improvement gives the group a grounded reason for hope: confidence in its capacity to turn errors into information.

From Exchange to Stewardship

The handoff is often treated as the edge of responsibility: my work ends when yours begins. Cross-functional maturity reframes it as overlapping stewardship. The sender remains responsible for intelligibility; the receiver becomes responsible for signaling the conditions required to carry the work; both attend to the person or public affected beyond them. This arrangement clarifies ownership while making the boundary relational.

Such stewardship requires concrete support. People need time to prepare and confirm consequential transfers. Recurrent bridge work needs recognition and backup; boundary spanning consumes attention (Marrone, 2010). Escalation routes must exist when reciprocal accommodation reaches its limit. A structurally impossible handoff requires organizational redesign because individual care has finite reach.

Within those limits, the exchange offers a compact practice of organizational life. It joins specialized strengths while preserving difference, makes a clear signal answerable, turns concern into usable support, and treats improvement as a sequence of observable adjustments. In the space between "sent" and "received," a group learns whether interdependence is merely declared or genuinely held.

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